Functions & Transformations

Section 1: Function Notation, Domain & Range

1 If $f(x) = 3x^2 - 2x + 5$, what is the value of $f(-2)$?

a. 13

b. 17

c. 21

d. 25

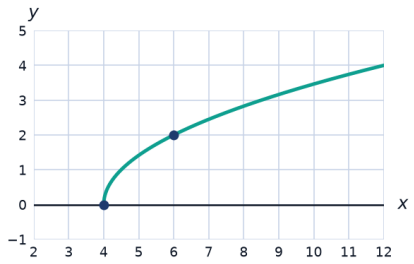
2 What is the domain of the real-valued function $g(x) = \sqrt{2x - 8}$?

a. $x \geq 4$

b. $x > 4$

c. $x \leq 4$

d. $x \geq 0$



3 The function h is defined by $h(x) = \frac{7}{x-3} + 4$. For what value of x is $h(x)$ undefined?

a. -3

b. 0

c. 3

d. 4

Section 2: Function Transformations

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4 The graph of $y = f(x)$ is shifted 4 units to the right and 3 units down. Which equation represents the transformed function $g(x)$?

a. $g(x) = f(x - 4) - 3$

b. $g(x) = f(x + 4) - 3$

c. $g(x) = f(x - 4) + 3$

d. $g(x) = f(x + 4) + 3$

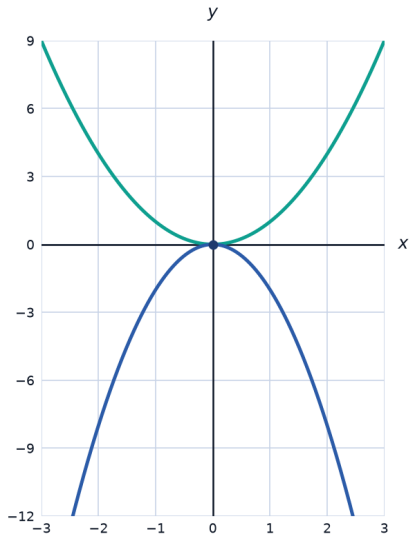
5 If the graph of $f(x) = x^2$ is reflected across the x-axis and vertically stretched by a factor of 5, what is the resulting equation?

a. $y = -5x^2$

b. $y = 5(-x)^2$

c. $y = -\frac{1}{5}x^2$

d. $y = -(5x)^2$



6 The function $f(x)$ has a local minimum at $(2, -5)$. What is the coordinates of the corresponding local minimum on the graph of $y = f(x + 3) + 4$?

a. $(-1, -1)$

b. $(5, -1)$

c. $(-1, -9)$

d. $(5, -9)$

Section 3: Composite & Inverse Functions

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7 If $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, what is the value of $f(g(3))$?

a. 17

b. 19

c. 21

d. 25

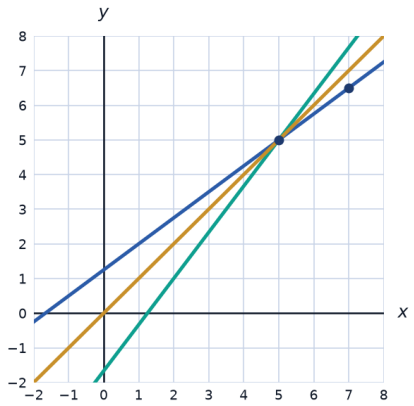
8 The function $f(x) = \frac{4x-5}{3}$ has an inverse function $f^{-1}(x)$. What is the value of $f^{-1}(7)$?

a. $\frac{13}{4}$

b. $\frac{19}{4}$

c. 5

d. 6



9 If $f(x) = 3x - 4$ and $g(x) = x + 2$, for what value of x does $(f \circ g)(x) = (g \circ f)(x)$?

a. -1

b. 0

c. 1

d. All real numbers

Section 4: Contextual Function Interpretation

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10 The temperature $T(m)$ of an oven m minutes after being turned on is modeled by $T(m) = 70 + 25m$ for $0 \le m \le 15$. What does the number 70 represent?

a. The initial temperature of the oven in degrees Fahrenheit.

b. The maximum temperature of the oven.

c. The rate at which the oven heats up per minute.

d. The time required to reach operating temperature.

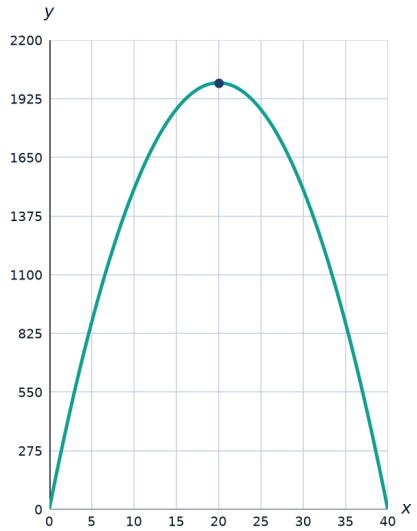
11 A company models its total revenue $R(p)$ from selling a product at price p dollars by $R(p) = -10(p - 25)^2 + 6250$. What price p maximizes the revenue?

a. 15

b. 20

c. $25 dollars$

d. 30 dollars



12 The volume $V(t)$ in gallons of water remaining in a pool being drained is given by $V(t) = 4500 - 30t$, where t is in minutes. How many minutes will it take for the pool to be completely empty?

a. 120

b. 150

c. 180

d. 200

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